

Three Kinks and a Null: Formal Trend-Break Tests on the Public Record of AI Progress

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July 2026

analysis pass of record: 2026-07-16, `natex` v0.2.0

Abstract

Claims that AI progress “bent” at some dated event — a model release, an export-control round, a regulation — are usually established by eye. We subject four such claims to formal regression kink designs in calendar time and a difference-in-kinks (DiK) design, using the `natex` v0.2.0 estimators (following Böckerman, Jysmä, and Kanninen, 2025) on Epoch AI’s public datasets, with a mandatory falsification battery: bandwidth–donut sensitivity grids, shifted-cutoff placebo-kink grids, density and covariate kinks, and a sibling-series falsification. Three kinks and one null survive. GPQA-Diamond scores kink at the o1-preview release (slope roughly $2.4\times$, placebo empirical size 0/7 — the bend localizes to the date). The METR 50% time-horizon slope nearly triples, but pre-side placebos also reject: a credible reasoning-era bend whose attribution to 2024-09-12 fails. The Epoch Capabilities Index, run as the guard against “everything kinked in 2024,” is a clean null at the same date in a series where the test demonstrably has power. And China’s legal AI-chip stock bends down by ≈ 0.56 log-units per year relative to hyperscaler controls at the October 2023 export controls, with the October 2022 round as a passing placebo and a smuggling-attenuated total-stock estimate about half the size and fragile. A graveyard of rejected designs — a $t = 12.9$ “kink” that fails every placebo, a sign-unstable Chinchilla null, and an EU AI Act threshold that is a step with bunching rather than a kink — illustrates why eyeballed trend breaks need placebo grids.

1 Introduction

The public conversation about AI progress is full of dated trend breaks. Reasoning models “changed the slope” in late 2024; export controls “bent” China’s compute accumulation; the EU AI Act “chilled” large training runs. These claims are almost always established by drawing a line through a scatter plot and pointing at a date. Yet a slope change at a known cutoff is precisely the estimand of the regression kink design (RKD), and a slope change for one group relative to another at the same cutoff is the estimand of the difference-in-kinks (DiK) design formalized by Böckerman, Jysmä, and Kanninen [1]. Treating eyeballed breaks as formal kink candidates buys three things: an explicit identifying assumption, a standard error, and — most importantly — a falsification battery that can *reject* the claim.

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This paper runs that battery on four trend-break claims drawn from Epoch AI’s public datasets [3]. The tooling is `natex` v0.2.0 [4], an open-source natural-experiment toolkit in the lineage of automated discontinuity-design discovery [2], whose kink module implements sharp and fuzzy RKD and DiK estimators following [1]. We emphasize that this is known-cutoff candidate *evaluation*, not unknown-kink discovery: every cutoff below is externally dated (a model release, a regulation date), and none was searched for. Searching for cutoffs would require search-calibrated selective inference beyond the scope of both the source paper and this exercise.

Why do eyeballed breaks need placebo grids? Because a calendar-time kink test asks a sharper question than the eye does. A series can genuinely accelerate over an *era* while providing no evidence that the bend happened at any particular *date*; conversely, a smoothly super-exponential series will show a nominally enormous “kink” at every date one tests. The shifted-cutoff placebo grid separates these cases: a significant kink at the true cutoff alongside significant kinks at shifted cutoffs means bend existence without date attribution, and rejections everywhere mean curvature masquerading as a kink. Our four headline results span this taxonomy almost perfectly — one date-localized kink, one era bend that fails date attribution, one clean null run deliberately as a falsification guard, and one difference-in-kinks with a passing placebo round — and the graveyard (Section 5) shows each failure mode in the wild.

2 Data

All series are public Epoch AI datasets (CC-BY 4.0, <https://epoch.ai/data>) [3], retrieved July 2026; the analysis pass of record is dated 2026-07-16 and used `natex` v0.2.0. No data files ship with this paper; the figure pipeline (`paper/figures/make_figures.py` in the repository) reads a local extraction of the public CSVs, recomputes every headline estimate, and asserts it against the numbers of record before drawing.

GPQA-Diamond. Benchmarking-Hub series `gpqa_diamond`: 180 dated models (45 pre / 135 post cutoff), sample starting 2023-03-14. Primary outcome is `logit(mean score)`: the score ceiling at 1 bends raw-score slopes mechanically near the top, and the logit removes that artifact. GPQA is the graduate-level science QA benchmark of [6].

METR 50% time horizon. `metr_time_horizons_external` (METR’s long-task suite [5] as mirrored by Epoch): 48 dated models (12 pre / 36 post). Outcome is $\log_2$ of the 50%-success time horizon in minutes.

Epoch Capabilities Index (ECI). `epoch_capabilities_index`, an IRT-linked aggregate capability index; 356 dated, scored models. Used as the sibling-series falsification.

AI chip stocks. `ai_chip_owners/cumulative_by_designer`: quarterly cumulative compute stocks in H100-equivalents by owner. Outcome is $\ln(\text{cumulative H100e})$; treated series China, control series the hyperscalers (Amazon + Microsoft + Google + Meta). The 2026-03-31 quarter is flagged `Incomplete` and its “cumulative” totals *decrease*; it is dropped.

3 Methods

3.1 Estimators

Let x be the running variable centered at the cutoff and let $m[s]$ denote the one-sided derivative of $E[Y \mid x]$ at zero on side s . `natex` defines every kink right-minus-left,

$$\kappa = m[\text{right}] - m[\text{left}], \quad (1)$$

estimated by local polynomial fits in which each side (and each group cell for DiK) receives its own intercept and slope, weighted by a kernel in $|x| \leq \text{bandwidth}$ (the weighted objective of [1], Equation 8). Defaults throughout: local linear fits, triangular kernel; uniform kernel, second-degree fits, and donut exclusions appear as explicit sensitivity choices. The sharp DiK estimand contrasts two such kinks,

$$\tau_{\text{DiK}} = (\kappa_{\text{treated}} - \kappa_{\text{control}}) / \Delta, \quad (2)$$

with Δ the known policy-kink change. In the China design the paper’s pre/post *periods* are aliased to control/treated *groups* at a common calendar cutoff: the export-license schedule bends for China only, so the Böckerman contrast is taken across groups and the controls difference out the global supply bend. Identification is parallel kinks: absent the controls, China’s growth bend would have matched the hyperscalers’.

All calendar-time runs use a dummy unit denominator ($\Delta = 1$, the `--policy-kink 1.0` convention), so τ is a *descriptive slope change* in outcome units per day, not a marginal causal response to a measured policy variable.

3.2 Time as the running variable

The governing caveat, quoted verbatim from the `natex` method card, `docs/method_cards/kink.md`:

Calendar-time RKD — “did the trend bend at this dated event?” — is a legitimate use of the estimator but **not** a manipulable-running-variable design: no unit sorts itself across a date, so density/manipulation arguments give no protection. Identification reduces to a single untestable assumption: *no co-located slope-changing event* — nothing else may bend the outcome’s expected slope at that exact date.

Two practices are therefore mandatory rather than optional. First, always run the shifted-cutoff *placebo-kink grid* and read it as separating bend existence from date attribution: a significant kink at the true cutoff plus significant kinks at shifted cutoffs means the series bends over an era, not at the event. Second, run a *sibling-series falsification*: an aggregate or related series in which the same test demonstrably has power but reads null at the candidate date. Sections 4.2 and 4.3 show both practices doing real work.

3.3 Inference

Standard errors are HC1 sandwich by default, with the degrees-of-freedom correction applied *jointly* across all side/period cells ($n - k$ counts every observation and coefficient of the stacked regression). Cross-checks fitting each side separately with per-side $n - k$ reproduce the point

Series (source)	Design / cutoff	Headline (HC1)	Verdict
GPQA-Diamond, logit(mean score), 180 models	sharp RKD-in-time at o1-preview, 2024-09-12	+0.00258/day, se 0.00078, $t = 3.32$ (bw 540, tri)	credible kink, date-localized
METR 50% time horizon, log ₂ minutes, 48 models	sharp RKD-in-time at o1-preview, 2024-09-12	+0.00601/day, se 0.00282, $t = 2.13$ (bw 720, tri)	credible era bend; date attribution fails
Epoch Capabilities Index, all models	sharp RKD-in-time at o1-preview (falsification)	−0.00725 pts/day, se 0.00895, $t = −0.81$ (bw 540, tri)	clean null — the guard the other two need
China vs. hyperscaler ln cu- mulative H100e stock	sharp group-DiK at ex- port controls, 2023-10-17	−0.00154/day, se 0.00047, $t = −3.30$ (bw 548, tri)	credible kink with a magnitude hon- esty band

Table 1: The four formal results of the pass (**natex** v0.2.0, HC1 standard errors). “tri” = triangular kernel; bandwidths in days.

estimate exactly and report SEs roughly 5% larger at $n \approx 50$: in the METR cross-validation (uniform kernel, bandwidth 720), **natex** and an independent per-side OLS fit agreed on the kink to 10 decimals (both 0.0067291763) with SEs 0.0023236 (joint-cell) versus 0.0024334 (per-side) — a convention difference, not an error. Cluster-robust CR1 covariance with $t(G-1)$ critical values is used where clustering is stated. Wald intervals are conventional local-polynomial inference and may retain smoothing bias; degree-2 and bandwidth sensitivity are part of the required battery. Failed computations return NaN, never zero. The diagnostics battery — bandwidth×donut sensitivity grids, placebo-kink grids with empirical size, density kinks, covariate kinks — follows the validation figures of [1] as implemented in **natex.kink**.

4 Results

Table 1 summarizes the four designs. All candidates had externally dated cutoffs; none was searched for.

4.1 GPQA-Diamond at o1-preview: a date-localized kink

The logit-score slope changes from 0.00188 to 0.00446 logit-units per day at the o1-preview release — in raw terms roughly $15 \rightarrow 34$ percentage points per year, a $\sim 2.4\times$ acceleration (Figure 1). The headline estimate is +0.00258 per day (se 0.00078, $t = 3.32$; bandwidth 540, triangular kernel, cells 44/111). The battery is uniformly supportive: the estimate is positive in 8/8 bandwidth×donut cells (t from 2.15 to 3.71), and the donut *strengthens* it, ruling out single-point dependence at the cutoff; the placebo-kink grid has empirical size **0/7** — no shifted cutoff rejects; a McCrary-style release-date density kink is null ($t = 0.76$), so the score bend is not an artifact of a release-frequency bend; a training-compute covariate kink is null ($p = 0.54$); and a degree-2 fit keeps the sign and magnitude at three times the standard error. This is the cleanest result of the pass: the bend localizes to the date.

The standing caveat belongs in every sentence of interpretation: composition — reasoning models entering the release stream — *is* the mechanism, so τ is a property of the release stream,

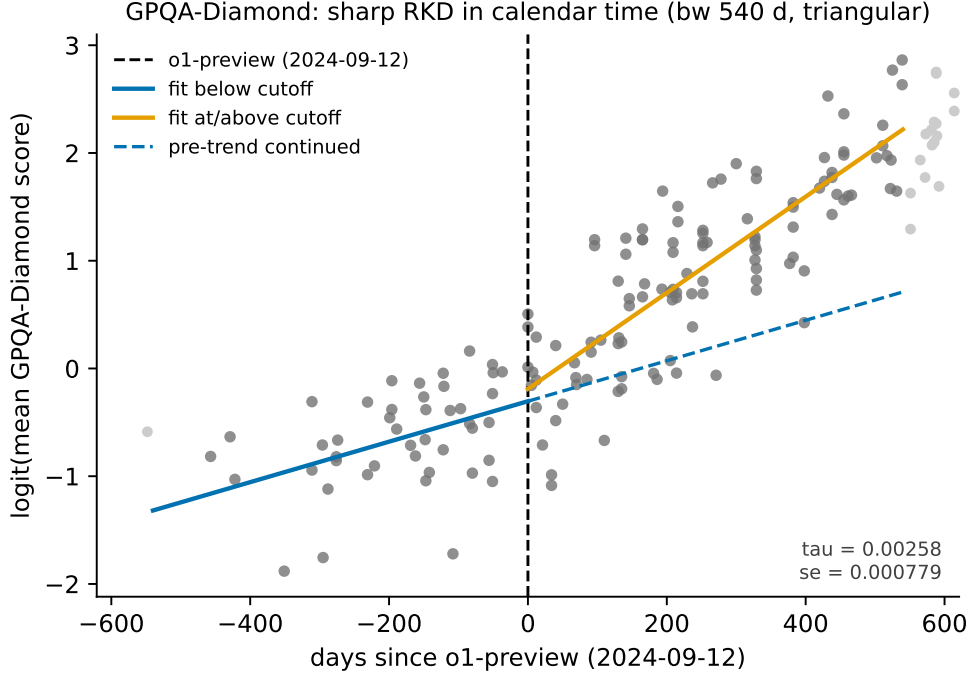


Figure 1: GPQA-Diamond, $\text{logit}(\text{mean score})$ against days since o1-preview (2024-09-12). Solid lines: kernel-weighted local-linear fits on each side (bandwidth 540 days, triangular); dashed: the pre-cutoff trend continued. The slope roughly $2.4\times$'s at the cutoff, and the placebo grid (empirical size 0/7) localizes the bend to the date.

not a statement that individual models improved. Benchmark contamination drifting over time remains an unmodeled confounder.

4.2 METR time horizon: a real bend that cannot be dated

The $\log_2$ 50%-time-horizon slope rises from 0.00331 to 0.00932 per day at bandwidth 720 (triangular): a doubling time of 9.9 months before the cutoff falling to 3.5 months after (Figure 2). The headline kink is $+0.00601$ (se 0.00282, $t = 2.13$); the full-sample uniform-kernel fit gives $+0.00491$ (se 0.00095, $t = 5.17$; doubling time $7.6 \rightarrow 3.6$ months). The estimate is positive in 8/8 bandwidth $\times$ donut cells (0.0044–0.0085), and the 80%-horizon variant agrees ($+0.00454$, $t = 4.08$), though with only three pre-cutoff points it is directional corroboration only.

But the placebo grid smears on the pre side: at bandwidth 720, shifted cutoffs at -270 , -180 , and -90 days all reject ($p = 0.030, 0.014, 0.004$) with estimates the size of the headline kink, while post-side shifts are clean nulls (empirical size 3/6). With 11–12 pre-cutoff models, adjacent placebo windows share most of their data, and the design cannot distinguish 2024-09-12 from any date within roughly ± 270 days on the pre side. The honest verdict, quoted from the analysis of record: *“credible slope change with failed date-localization — report as reasoning-era slope doubling, not ‘o1-preview caused X’.”* This is the era-bend/event-bend contrast with Section 4.1 in a single dataset pair.

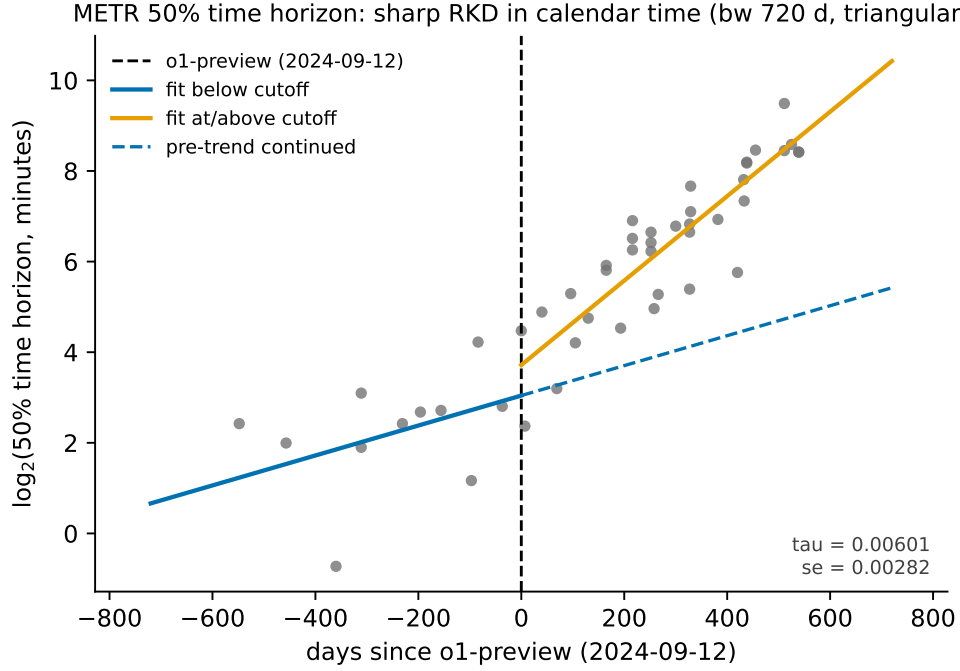


Figure 2: METR 50% time horizon ($\log_2$ minutes) against days since o1-preview. The slope nearly triples (doubling time $9.9 \rightarrow 3.5$ months at bandwidth 720), but pre-side placebo cutoffs also reject: an era bend, not a dated event.

4.3 The Epoch Capabilities Index: a null with power

If “everything kinked in 2024” — a global measurement or composition shift — then per-benchmark kinks would be uninformative. The ECI falsification run was launched *expecting* a null, and it delivered one: at the o1 date the all-models kink is null in 8/8 grid cells (t from -0.60 to -1.54 ; headline -0.00725 points/day, se 0.00895), with the index gaining roughly 18 points per year on both sides of the cutoff (Figure 3). Crucially, the same series’ placebo grid rejects at 4/7 shifted cutoffs — the test demonstrably has power in this data, from composition-churn bends elsewhere in the series — and still reads zero at 2024-09-12. The instrument works; the dial reads zero. The per-benchmark bends of Sections 4.1–4.2 are therefore not a global measurement artifact. This is partly by construction (IRT linking smooths regime shifts) and partly because o1-era gains concentrate in reasoning-heavy benchmarks.

Two recorded lessons ride along. First, an analyst-script frontier filter — `cummax().diff().fillna(1) > 0` over a series with 231 NaN scores — manufactured 347 false “frontier records”; the clean frontier has 25 points and is *not* an estimable design — four left-cell points, failing donut, bandwidth-direction, and placebo checks in every direction — and should never be quoted. Second, a side finding: the training-compute covariate kink among ECI-listed models is -0.00265 $\log_{10}$ FLOP/day at the o1 date ($p = 0.021$), corroborating the “pretraining-compute frontier stalls” narrative without itself being a design.

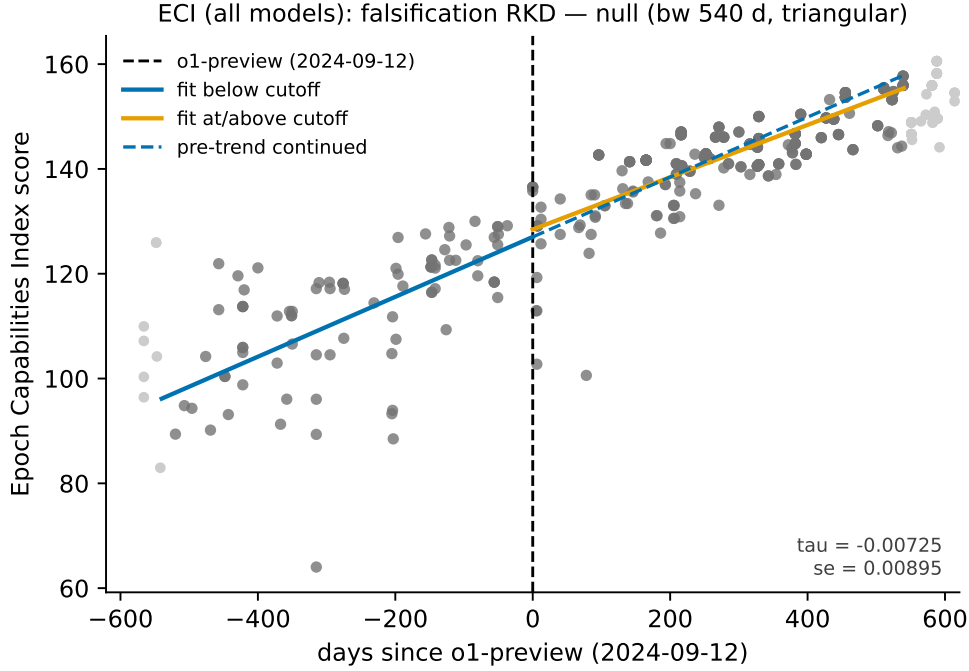


Figure 3: Epoch Capabilities Index (all models) against days since o1-preview. The two-sided fit and the continued pre-trend nearly coincide: a clean null ($t = -0.81$) in a series where the placebo grid shows the test has power (4/7 shifted cutoffs reject). The guard the per-benchmark kinks need.

4.4 China’s chip stock at the October 2023 export controls

The only true difference-in-kinks structure in the corpus: the US export-license schedule bends for China only, at a common dated cutoff (2023-10-17), so treated = China and control = hyperscalers (Amazon + Microsoft + Google + Meta), and the controls difference out the global supply bend (their own kink: $-0.00067/\text{day}$). The legal-stock DiK is -0.00154 log-units per day (se 0.00047, $t = -3.30$; bandwidth 548, triangular), i.e. a growth-rate change of roughly **-0.56 log-units per year** (Figure 4). By-year CR1 clustering keeps significance ($t = -5.15$) with the explicit few-cluster caveat ($G = 4$). The estimate is negative in **25/25** specifications.

Falsifications behave: the October 2022 first-round placebo — the round whose A800/H800 compliance loophole predicts no bite — is null/positive ($+0.0018$, $p = 0.15$), and the placebo-treated group (“Other” vs. hyperscalers) is clean at bandwidths 548–730 while rejecting at 365, so the defensible range is bandwidth 548–730 with DiK -0.0009 to -0.0015 . The honesty band: the *total-stock* series including smuggled chips gives a DiK roughly half the size and fragile (-0.00076 , $t = -1.54$ at bandwidth 548) — the smuggled series, which starts 2024-03-31 and is itself a treatment response, substitutes for lost legal supply. Serial dependence in a six-points-per-cell cumulative series makes every $|t|$ optimistic; the sign stability is the real evidence. The verdict of record: *“the policy bent the legal channel ≈ -0.3 to -0.56 ln-units/yr; the effect on China’s actual compute stock is smaller and fragile.”*

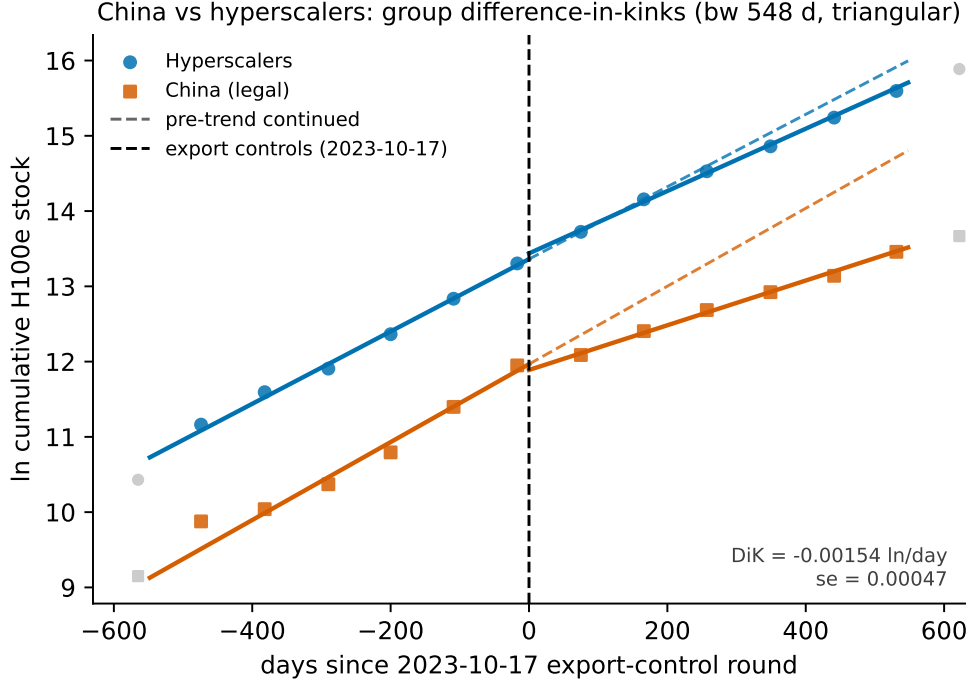


Figure 4: Group difference-in-kinks: ln cumulative H100e stock for China (legal) and the hyperscaler controls against days since the 2023-10-17 export-control round (bandwidth 548 days, triangular). Both series bend at the cutoff; the DiK is their difference, -0.00154 log-units/day ($\approx -0.56/\text{yr}$).

5 The graveyard

Rejections are results. Three claims that look at least as impressive as Table 1 in a scatter plot died in the battery.

Datacenter growth “kinked at ChatGPT.” The GPU-cluster series (ln cumulative H100e by first-operational date) shows a nominal kink at ChatGPT’s release with $t = 12.9$ — the largest t -statistic of the entire exercise. Its placebo grid rejects at **7/7** shifted cutoffs (empirical size 1.0): a smoothly super-exponential cumulative series bends *everywhere*, and HC1 inference on serially dependent cumulative data is meaningless. The design was discarded; the estimate is descriptive curvature, not a kink. (The dataset variant named in the original claim, datacenter power timelines, was infeasible outright: one pre-ChatGPT observation.) This is the cautionary tale for every “the curve bent at [event]” plot drawn on cumulative infrastructure data.

Chinchilla. Tokens-per-parameter among language base models at the Chinchilla paper’s release date (2022-03-29): kink $t = -0.91, -0.05, +1.48$ at bandwidths 365, 540, 730 — a sign-unstable null. Even a genuine regime change (prior work shows a two-to-three-month adoption ramp) does not produce a slope kink dateable to the paper’s release.

The EU AI Act threshold is a step with bunching, not a kink. The Act’s systemic-risk presumption at 10^{25} training FLOP imposes a *level* discontinuity in obligations, not a marginal-rate kink, so RKD/DiK is the wrong estimand shape. Worse for any kink or RD design at that cutoff, the running variable sorts: post-Act, the model-density just above the line is depleted (below/above contingency: Fisher exact OR = 0.17, 95% CI 0.05–0.62, $p = 0.0071$; roughly 83% of the expected above-line mass missing), which violates any no-sorting requirement. Kink-shaped probes confirm nulls (e.g. an open-weights-share DiK of -0.93 , se 0.72). The bunching test remains the right tool at this threshold — and the bunching itself, an avoidance response specific to the statutory line and the post-Act period, is the substantive finding.

6 Discussion

Four lessons generalize beyond these datasets.

Era bends and event bends are different claims. The METR and GPQA series tell almost the same visual story — capability slopes steepen around late 2024 — yet the placebo grid cleanly separates them: GPQA’s bend localizes to the o1 date (size 0/7) while METR’s does not (pre-side size 3/6). Public discussion rarely distinguishes “the trend changed in this era” from “this event changed the trend”; the shifted-cutoff grid makes the distinction mechanical.

Nulls need power certificates. The ECI falsification is informative precisely because the same series rejects at 4/7 shifted placebo cutoffs: the test has power in this data, and still reads zero at the candidate date. A null from an underpowered test would have guarded nothing.

Report honesty bands, not point claims. The China result is strongest as a band: legal-channel bend ≈ -0.3 to -0.56 log-units/yr across the defensible bandwidth range, total-stock effect roughly half and fragile because smuggling — itself a response to the policy — substitutes for legal supply. Similarly, GPQA’s τ is a release-stream property with composition as the mechanism, not per-model improvement.

The graveyard is the argument. The most significant statistic of the exercise ($t = 12.9$) belongs to a design that fails every placebo. Any workflow that stops at “the kink is significant” would have led with it. On the public record of AI progress, where series are short, serially dependent, and composition-churned, the falsification battery is not a robustness appendix — it is the analysis.

Limitations. Calendar-time identification rests on an untestable no-co-located-event assumption; placebo grids probe but cannot certify it. Several series are short (6 quarterly points per cell in the DiK; 11–12 pre-cutoff models for METR), making serial dependence corrections coarse and $|t|$ optimistic. Epoch’s compute figures are partly estimates, and benchmark contamination drifts over time. All results are properties of the *public record* of AI progress — release-stream composition included — not of any individual system.

Reproducibility

Estimates were produced with the open-source `natex` v0.2.0 kink module [4] on public Epoch AI data [3]; the per-design analysis records (inputs, CLI outputs, diagnostics JSON) and the case-study write-up of record live in the `natex` repository under `docs/case_studies/` (`epoch-kinks.md`). The four figures regenerate deterministically from the public CSVs via the committed script `paper/figures/make_figures.py`, which asserts every headline number against the case study before drawing.

References

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